



Kondaguduru Sai kumar

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M.Tech - Power Engineering

Electronics and Electrical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/10.0	Year
M.Tech	Indian Institute of Technology, Guwahati	9.32 (Current)	2025-Present
B.Tech	Guru Nanak Institute of Technology	7.63	2018-2022
Senior Secondary	Board of Intermediate Education	9.48	2016-2018
Secondary	Board of Secondary Education	9.5	2016

EXPERIENCE

- Indian Institute Of Technology Guwahati** July 2026 - Nov 2026
Teaching Assistant
 - Basic Electronics Lab.
 - Circuit Theory Course.

PROJECTS

- FPGA-Controlled 4-Leg DSTATCOM for Data Centers** Jul 2026 - Present
M.Tech Project under Assistant Prof. Sreenath JG, Dept. of EEE, IIT Guwahati
 - Formulated a Discrete Kalman Filter (DKF) control strategy for a 4-leg DSTATCOM to dynamically mitigate triplen harmonics, neutral wire accumulation, and sub-cycle voltage sags in high-density data center microgrids.
 - Designed the digital control architecture using pipelined Verilog RTL to execute fixed-point state estimations and multi-channel PWM on an FPGA, achieving sub-2-microsecond execution latency to outperform conventional DSP.
- RTL-to-GDSII Implementation of a 2x2 Systolic Array** July 2026
Self Project Github
 - Designed and implemented a 2×2 systolic-array architecture for matrix multiplication in Verilog, followed by RTL synthesis using the UMC 65nm standard-cell library in Synopsys Design Compiler.
 - Executed the complete physical design flow in Cadence Innovus, covering floorplanning, placement, CTS, routing, STA, and GDSII generation; performed post-layout simulation using Synopsys VCS.
- Multi-Domain 1×3 Network Router with CDC Synchronization** May 2026
Self Project Github
 - Designed a Verilog RTL-based 1×3 packet router with integrated asynchronous FIFOs, managing clock-domain crossing (CDC) between a 125 MHz source and three independent destinations (156.25, 100, and 50 MHz).
 - Developed a Verilog testbench to verify full router and FIFO functionality including Gray code pointer synchronization and independent packet delivery—and performed RTL synthesis using Intel Quartus Prime Lite.
- AHB-to-APB Bridge Protocol Verification using UVM** April 2023
Maven Silicon Github
 - Developed a UVM-based verification environment for an AHB-to-APB Bridge with reusable AHB/APB agents, monitors and scoreboard for transaction-level verification.
 - Verified AHB-to-APB read/write transactions with 26 transactions checked through the scoreboard, along with functional and cross coverage to ensure proper verification of the bridge.

TECHNICAL SKILLS

- HDL/Programming:** Verilog-HDL, C
- Tools:** ModelSim, Quartus Prime, Xilinx Vivado, Cadence Virtuoso*, MATLAB * Elementary proficiency

KEY COURSES TAKEN

- NPTEL:** Multi-Core Computer Architecture. July 2026 - Present
- EICT, IIT Roorkee:** CMOS Digital VLSI Design. July 2026 - Aug. 2026
- Advanced VLSI Design and Verification, Maven Silicon:** Verilog-HDL, System Verilog, UVM. May 2023

EXTRA CURRICULAR ACTIVITIES

- Semiconductor Workshop,** Attended Semiconductor IC Design and Manufacturing workshop by NEST cluster 2025.
- NIELIT,** RTL, IPIntegration and SoC Signoff June 2026 - July 2026
- Hobbies,** Badminton, Swimming, Yoga, Pranayama